■ LITHIUM OPERATIONS PLATFORM

Autonomous Operations Intelligence Desk & Digital Twin Engine

PS 5 — Resonance 1.0 | SCOPE, VIT Chennai
Status: **Phase 1 Complete** | DB: **SQLite 3 WAL**
Confidential — Feature Specification & Roadmap Document

1. System Architecture Overview

Layer	Technology Stack	Core Responsibility & Architecture Role
Frontend UI	Vanilla JS, Tailwind CSS, HTML5 Canvas	Reactive SRE dashboard, dynamic topology graph, time-horizon scrubber, and live SSE stream.
Backend API	FastAPI, Uvicorn, asyncio, Python 3.13	Real-time telemetry engine, chaos injector, safety kill-switch, and REST API.
Database	SQLite 3 with WAL (Write-Ahead Logging)	High-concurrency persistence for incidents, immutable audit ledgers, and post-mortems.
ML & AI	IsolationForest + Google Gemini 2.5	Telemetry anomaly scoring with heuristic fallbacks and automated senior SRE PIR reports.

2. Complete Feature Matrix (Currently Implemented & Live)

Feature Name	Description & Engineering Implementation	Status
1. Real-Time System Health & KPIs	Instant metrics for Overall Health (99.8%), Microservices Online (5/5), P99 Latency (32ms), and MTTR (16.2s) with live UTC clock and API connectivity badge.	**ACTIVE**
2. Microservice Dependency Topology	Custom HTML5 Canvas directed graph (Clients ➔ Gateway ➔ Auth ➔ Payments ➔ SQLite DB) with dynamic node coloring (Green/Red/Amber/Blue) and click drawer.	**ACTIVE**
3. Chaos Injector & Outage Presets	Instant triggers for Database Deadlock (DB_LOCK), Auth Memory Leak (OOM Kill), and DDoS Traffic Spikes with durable SQLite storage and dynamic blast radius updates.	**ACTIVE**
4. Live Server Telemetry Controller	Interactive [Input Server Data] modal allowing operators to customize the failing service, scrub exact Calculated Risk Rate (1–100), latency, and test playbooks.	**ACTIVE**
5. Dual Autonomy Modes (Hybrid / Auto)	Hybrid Autonomous mode requires human operator approval ([Approve & Execute Fix]). Full Autonomous mode runs a 2.0s countdown and auto-remediates zero-touch.	**ACTIVE**
6. Safety Kill-Switch & Override	Backend security interlock (POST /api/system/kill-switch) that rejects autonomous AI modifications with HTTP 403 Forbidden while manual override is active.	**ACTIVE**
7. Digital Twin Time Scrubber (USP #1)	Scrub forward into the future (0m to +30m) to inspect projected saturation cliffs up to 12 minutes in advance, with a one-click preemptive fix action.	**ACTIVE**
8. Automated PIR & PDF Export (USP #2)	AI post-mortem generation via Google Gemini with causal flow diagram (Root Cause ➔ Failure ➔ Remediation ➔ Recovery) and direct browser PDF download.	**ACTIVE**
9. SQLite 3 WAL Persistence Layer	Durable ACID storage in backend/lithium.db with WAL mode, persisting all active/resolved incidents, post-mortems, and an immutable audit ledger with JSON export.	**ACTIVE**
10. Machine Learning Anomaly Detection	Backend scikit-learn IsolationForest model trained on baseline telemetry to output real anomaly scores and confidence percentages via /api/digital-twin/telemetry.	**ACTIVE**
11. Real-Time Telemetry Streaming (SSE)	Server-Sent Events stream (/api/v1/telemetry/stream) broadcasting live metrics every 1.5s to an interactive terminal with category filters (LOGS, METRICS, SECURITY, CLOUD).	**ACTIVE**
12. Single-Click Launch Automation	start.bat and run_app.bat scripts that verify environment, initialize SQLite WAL database, launch FastAPI on port 8000, launch web server on port 3000, and open the browser.	**ACTIVE**

3. Expansion Roadmap (What We Can Add Next)

These high-value feature concepts can be added next to maximize hackathon presentation points, impress judges, or prepare the platform for production deployment:

Proposed Feature	Technical Scope & Description	Demo & Judging Impact
1. AI SRE Copilot Chat Drawer	Slide-out interactive chat window where operators converse in natural language (e.g. *'Why did SQLite spike?'* or *'Roll back commit #8f3a'*), powered by Gemini AI with tool calling.	**Highest Impact** Demonstrates conversational agentic AI for DevOps.
2. Multi-Region / Cloud Switcher	Header dropdown allowing operators to toggle the operations desk between multiple cloud environments: AWS us-east-1, GCP europe-west3, and Azure Central India.	**High Impact** Gives the platform an enterprise, multi-cloud scale feel.
3. Simulated PagerDuty / Slack Webhooks	Notification toast module that triggers realistic webhook alerts into simulated Slack channels or PagerDuty on-call escalation policies when P1 incidents occur.	**High Impact** Proves integration with standard SRE workflows.
4. Financial Loss & Downtime Counter	Live dynamic ticker on the incident banner showing *'Estimated Downtime Cost Avoided: $1,420/min'*, dynamically calculating ROI saved by the self-healing engine.	**High Impact** Compelling business and ROI presentation for judges.
5. Custom Playbook Script Editor	Configuration view allowing SREs to upload or edit custom Bash/Python remediation scripts tied to specific microservice failure signatures.	**Medium Impact** Demonstrates flexibility for custom enterprise stacks.
6. Audio Sound Effects & Voice Alerts	Subtle audio cues: a sci-fi klaxon sound during incident injection and a positive audio chime when self-healing restores cluster health.	**Stage Presence** Captures immediate audience attention in live demos.

4. Quick Start Guide for Team Members

- **Zero Setup Needed:** All Python dependencies (FastAPI, Uvicorn, Scikit-learn, Google-GenAI, ReportLab) are installed.
- **Single-Click Run:** Double-click **start.bat** in the project root to launch both the backend API and frontend dashboard.
- **Backend Engine:** Running on **http://localhost:8000** (Interactive API Docs: **http://localhost:8000/docs**).
- **Frontend Dashboard:** Running on **http://localhost:3000** (Auto-connects to backend with live status badge).
- **SQLite Database:** Located at **backend/lithium.db** (WAL mode enabled, durable across restarts).